

May 17, 2026

Contents

1	Introduction	1
2	Project goal	4
3	Method	5
3.1	Dataset	5
3.2	Data Pre-processing	5
3.3	Feature Engineering	8
3.4	Models	9
3.4.1	Wide and Deep Model	9
3.4.2	MLP	12
3.4.3	Random Forest	12
3.4.4	SVM	13
3.5	Implementation	13
3.6	Class Imbalance	13
4	Deployment	15
5	Results	16
6	Discussion	19
7	Conclusion	21

Chapter 1

Introduction

Cervical intraepithelial neoplasia (CIN) is the second most common malignancy in women all over the world as the abnormal growth of cells on the surface of the cervix could induce cervical cancer [1] [2]. The clinical reference standard based on colposcopically directed biopsies divided them into various grades such as Grade1, Grade2, and Grade3 [3] [4]. The Grade1 CIN is called a Low-grade squamous intraepithelial lesion(LSIL) and is usually considered benign or non-cancer. Grade2 and Grade3 belong to a High-grade squamous intraepithelial lesion(HSIL) and is considered to be cancer [5]. Human papillomavirus (HPV) has been identified as the main cause of cervical cancer [6], Especially, HPV16 or HPV18 are the two most dangerous HPV genotypes in the Pap test [7] for detection of CIN2 or worse in high-risk HPV infection women. The presence of Human Papilloma Virus(HPV) type 16, 18 will lead to diagnostic of cervical cancers and pre-cancerous cervical lesions in 70 out of 100 patients [8].

Appropriately treatment of pre-invasive lesions (cervical intraepithelial neoplasia; CIN) identified in a systematic call and recall screening programs has decreased in incidence and mortality [9]. With a continuously evolving evidence base, developing a clinical decision support system is a currently unmet need to assist clinicians in using these new technologies to promote prevention. Today, the typical treatment for ASCUS and LSIL smears is still immediate referral to colposcopy. The decision to proceed with colposcopy immediately may overload the colposcopy clinic. Moreover, lead to over-treatment due to undetectable findings. Unnecessary colposcopy could be reduced through this support decision system.

There have been some works that had been carried out in predicting the

cervical cancer using machine learning models but not many for predicting cervical cancer given CIN tests. [10] went about the task of predicting cervical cancer by using a dataset of 145 patients with features that were majorly dominated by patients behavioural characteristics such as Number of sexual partners, Contraceptive usage, Marriage Age, e.t.c. [11] also developed predictive models for cervical cancer using 58 samples of gene expression profiling data consisting of 714 features, [12] did a survey on existing works done on the cervical cancer screening, this consisted of a lot of algorithms and also various datasets based largely on images and some numerical and categorical datasets. [13] works on CIN classification and Prediction, though it wasn't extended to cervical cancer prediction, they worked with a dataset of 508 instances with 122 features gotten from a questionnaire designed by the authors using various multi-class algorithms.

The prediction of cervical cancer is quite popular on the cervigram images with deep learning technology [14] [15] [16] [17]. Instead of using cervigram images, [18] developed a clinical decision support system (CDSS) for cervical cancer prevention and enhanced the identification of overdue patients their next routine/next screening and follow-up of abnormal results. [19] develop a data warehouse to support the management of the cervical cancer screening and then make decisions based on indicators. [20] and [21] present a decision support system based on an ensemble of Random Forests (RFs) to produce more accurate results. [22] develop a clinical decision support scoring system based deep neural network for personalized management of women with cervical abnormalities. All study as mentioned above in prediction only consider the TCT and HPV virus type while not the virus load; however, The role of viral load of high-risk HPV in the cervical screening has been reported [23] [24] [25] [26]. It has reached a consensus that the high viral load of HPV infection correlated with an increased risk of abnormality. However, one study shows that the virus load could not be reliable for the prediction of CIN2+ [27].

The data used in previous works is completely different from ours, the data that has been used in the previous work consist of a considerable amount of images datasets which uses Convolution Neural Network for prediction and in other works the attributes used are different such as marital status, number of sexual partners, smoking habits, alcohol consumption etc. We have also seen that usage of Random Forest trees, Support Vector Machine and also Multi-layer Perceptron gave some good results.

As part of this project our team builds a predictive model which is able

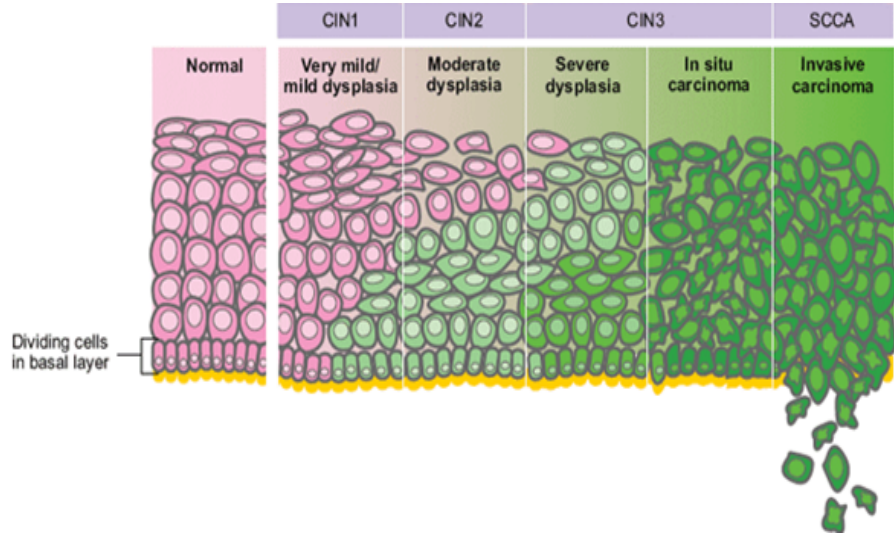


Figure 1.1: Types of CIN [30]

to predict if a patient has a cancer or not based upon the various attributes and their values which would save a lot of mundane human efforts. The given data set is labeled making it suitable for supervised learning.

This study combines the TCT, HPV virus type, and Virus load to be trained with several algorithms (SVM, Random forest, Multi-layer Perceptron, and Wide&deep learning [28]). Unlike the traditional algorithm, we applied the Wide&Deep model by jointly training a wide linear model (for memorization) alongside a deep neural network (for generalization) considering the Feature attributes, including both the categorical (Virus type) and numeric (Virus load) data. A viral load is an exponential number with a base of 10; We also evaluated the possibility of abnormality prediction using the maximum value of viral load in the later feature selection. The dataset consists of 5973 consecutive outpatients undergoing cervical cancer screening followed by colonoscopy testing at the gynecology department across 301 Hospital in China, collected from October 2016 to June 2018. The Synthetic Minority Oversampling Technique (SMOTE) [29] was applied to handle the imbalance issue in the dataset. The final results guide us to develop a webpage-based interface for cervical abnormality screening in clinical.

Chapter 2

Project goal

Cervical cancer is a leading cause of death among females claiming over 310,000 lives in 2018, especially in third world countries. Cervical intraepithelial neoplasia (CIN) is a growth of cells in the cervix that could cause cancer, early detection is extremely important for the successful treatment of cervical cancer, thus a clinical decision support system to aid early detection will be very helpful to minimise the death rate of cervical cancer patients. This study aimed to train a decision support system for clinical early Cervical cancer prediction with the Wide&Deep learning model. The given data-set consisted of 5973 consecutive outpatients undergoing cervical cancer screening followed by colonoscopy testing, collected across 301 hospitals in China from October 2016 to June 2018. The best results for overall-balanced recall and specificity for Wide&Deep model (92%, 88%, respectively) were achieved with Synthetic Minority Oversampling Technique(SMOTE). The sufficient sample size and promising model performance can pave the way for a user-centric web-based clinical decision-making system in the future. Our goal for this project involves the following listed below:

- A Clinical Support system for Cervical Cancer Detection
- Analysing the effectiveness of Wide and Deep Learning in comparison to Random Forest, SVM and MLP.
- Determining most important viruses for CIN detection using ML techniques.
- Deployment of a production ready webpage for CIN detection

Chapter 3

Method

3.1 Dataset

The data set consisted of 5973 consecutive outpatients undergoing cervical cancer screening followed by colposcopy testing collected from October 2016 to June 2018 across 301 hospitals in China. Histopathology results were taken as the gold standard of diagnosis.

The dataset consists of the test results of the HPV(Human Papilloma Virus), TCT(Thinrep Cytologic Test), different type of virus, and also the viral load of each type of virus along with the age of the patients ranging from 24 years to 84 years(Mean $\pm$ Std=41.9 $\pm$ 11.7). Total 30 attributes with 632 positive samples and more than 5000 negative samples are present in the dataset. The dataset has a sparsity of 92%. This dataset mainly concentrates on medical(clinical) tests such as HPV results and TCT results rather than behavioral features like smoking, sexual partners, marital status e.t.c. There are 4923 samples without CIN, 418 samples with CIN1, 546 samples with CIN2, and 86 samples with CIN 3 to be precise. The correlation matrix for all the features of dataset can be seen in the figure 3.1

3.2 Data Pre-processing

There is a total of 30 attributes. These are expanded to 39 after making age, and TCT categorized, and adding an extra feature that indicates the maximum value of viral load observed for each patient for easy processing. The dataset initially had four(4) classes(The three types of CIN(1-3) and no

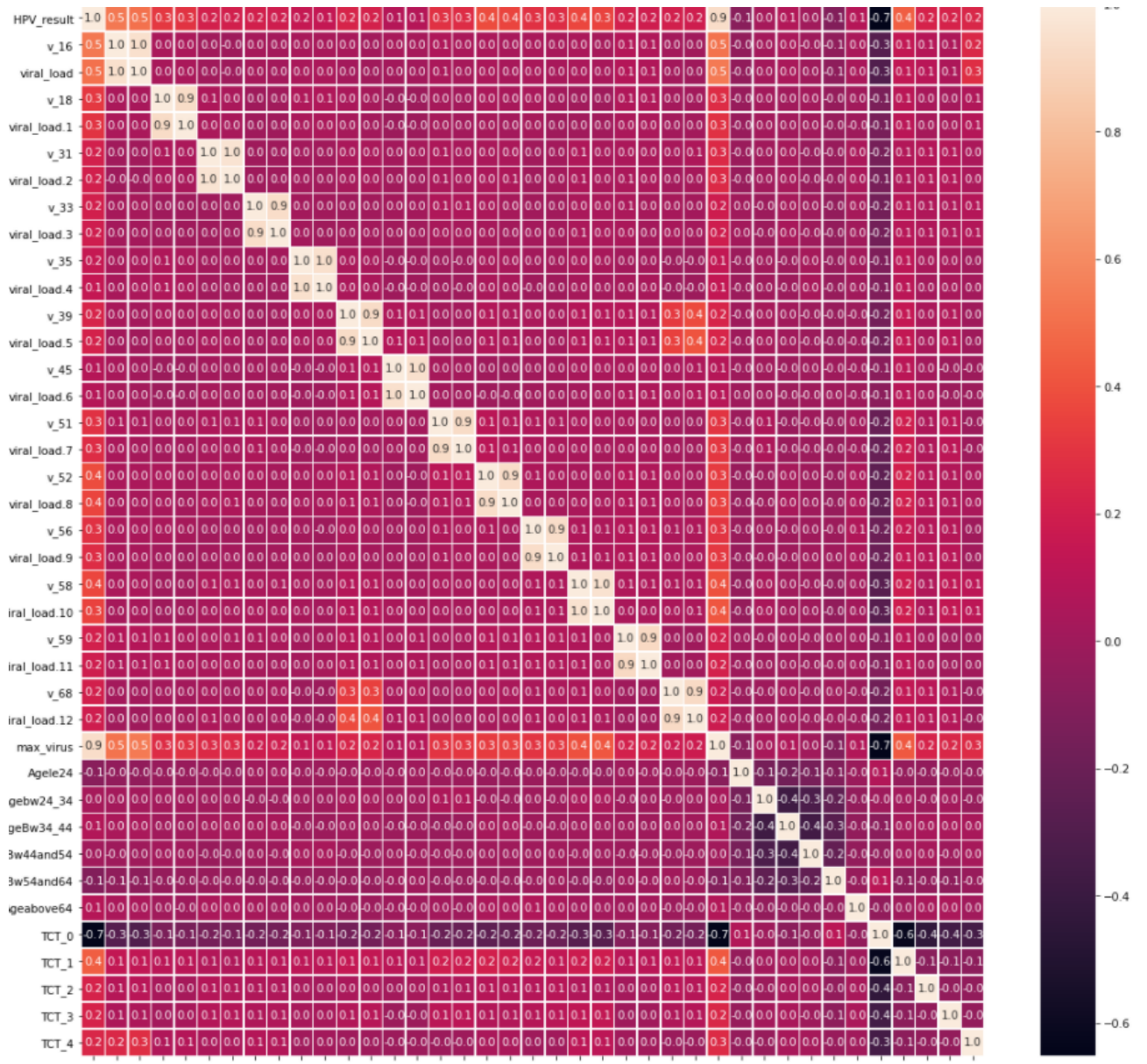


Figure 3.1: Correlation Matrix

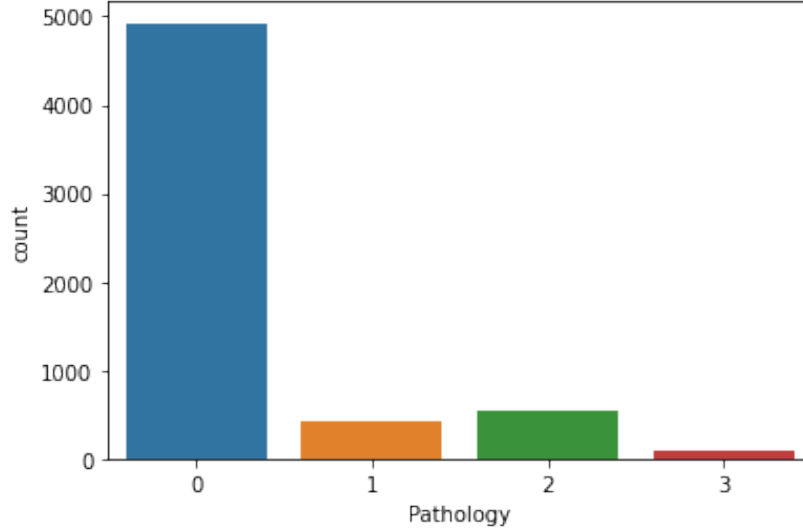


Figure 3.2: Class Distribution

CIN(0)) but has been reduced to two(2) classes for cancer detection with no CIN and CIN grade 1 representing No Cancer(Negative class) while CIN grade 2 and 3 representing the presence of cervical cancer(Positive class). Finally, there are 632 subjects with cancer and 5431 subjects without cancer in the total 5973 subjects.

Further details on the pre-processing of the features are as follows: Age was categorized as (<20; 21-29; 30-39; 40-49; 50-65; >65); For TCT, we observed that there are five values from 0 to 4(0: Normal,1: ASCUS, 2: ASC-H,3: LSIL, 4: HSIL+), dummy columns were created for this attribute namely TCT_0 to TCT_4 for each value 0 to 4 respectively, and each column would tell us if the each of the dummy column value is present or not(1/0); In case of viral-loads(for HPV DNA type:16,18,31,33,35,39,45,51,52,56,58,59,68), the maximum value in each type of virus is 10 million, so we have taken the power of each value for high computation efficiency, a new feature (max_virus) was created which represents the maximum virus load present for each patient . The Min-Max Scaler is used to normalize the data. We divided the columns into numeric, categorical, and categorical with identity for deep and wide learning. After preprocessing, there are total 39 attributes for 5973 samples,see Tab.3.2.

Table 3.1: Feature table

Feature Name	Value Type	Value Range
Age(bins)	Categorical	0/1
TCT(0-4)	Categorical	0/1
HPV Result(present or not)	Categorical	0/1
HPV Virus type	Categorical	0/1
HPV Virus Load	Numerical	0-7
MaxVirusLoad	Numerical	0-7
Pathology	Categorical	0/1

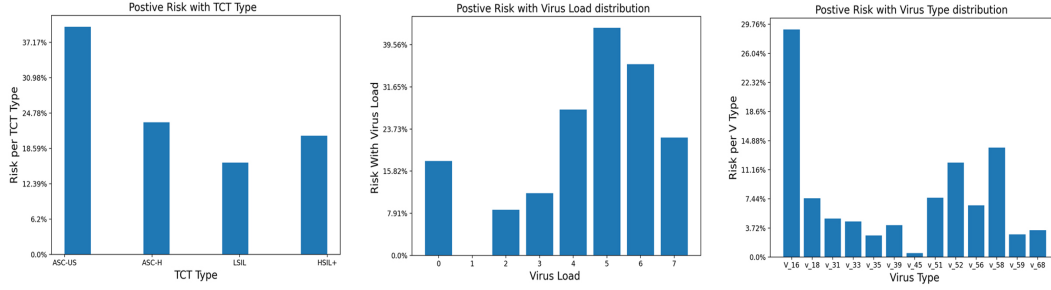


Figure 3.3: Positive risk with virus type distribution

3.3 Feature Engineering

In the traditional machine learning model, feature selection is a crucial part, which dramatically impacts the performance and stability of the model. In order to get insights into the feature distribution in the dataset, we visualized the distribution of the positive cases as in Fig. 3.3.

From Fig.3.3 right part, The distribution of positive patients with different virus types can be seen. v_16, v_52, v_58 has the highest risk in all the positive patients. From the previous report of CIN [31], the patients with v_16, v_18, v_31 or v_33 are under high risk. Hence we decided to use v_16, v_18, v_31, v_33 along with the virus types seen in the distribution: v_52, v_58.

Fig.3.3 middle one shows the distribution of virus loads of each type of virus. Here virus load means the number of each type of virus present in the patient. If the virus present in the patient is more than 10,000 in number then the patient will likely have cancer. But the fact that there is less number

Table 3.2: Hand-picked and Chi-square feature table

Feature Name	Value Type	Value Range
HPV Result(present or not)	Categorical	0/1
TCT- 0-4	Categorical	0/1
HPV Virus type-16,18,31,33,52,58	Categorical	0/1
HPV Virus Load-v16,18,31,33,52,58	Numerical	0-7
MaxVirusLoad	Numerical	0-7

of patients with less than 10,000 number of virus cannot be ignored. An attribute-based on this is called max-virus which gives the maximum number of the virus of whichever type of virus present in the patient. Moreover, If patients did not have any type of HPV then the patient’s TCT result might determine whether a sample is positive or negative.

We also used the Chi-Squared test for feature selection, where the top 15 features were selected given a Chi-score threshold. After the feature engineering, a new feature set was formed which consist of 19 attributes from the consistencies in the top 15 features of the Chi-squared test, medical facts, truth table distribution, and significant attributes from the feature visualization.

3.4 Models

To evaluate the performance of the wide and deep model, we compared its performance metrics with those of three other machine learning models which has been widely used for cervical cancer detection and prediction. These four models are discussed in this section.

3.4.1 Wide and Deep Model

The Wide&Deep model is a high-level framework developed by google research group for the recommendation of applications to users of the Google Playstore platform [28], which performed better than both (wide and deep) models individually. The basic idea is to combine the memorization power of wide models and the generalization power of deep neural networks to form a more powerful single model that handles the weakness of both models individually.

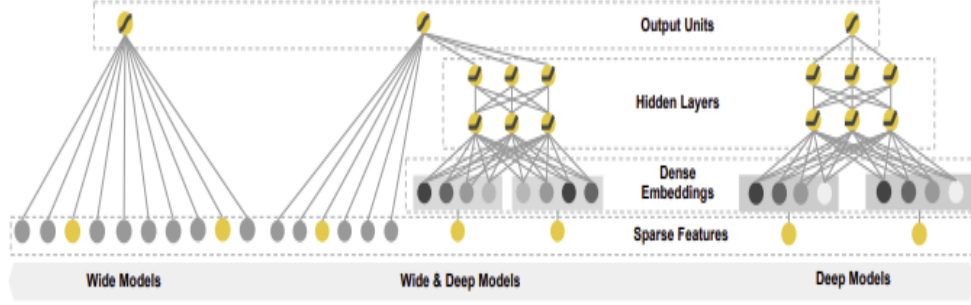


Figure 3.4: The spectrum of wide and deep model [28]

Unlike ensemble classifiers that train two or more classifiers individually and combine their predictions at the end, the wide&deep model is jointly trained, so the error from the output is back-propagated to both wide and deep parts simultaneously using gradient descent. All the model parameters are optimized simultaneously, making both parts act as a complement for each other, thus producing better performance.

Using only the wide model would memorize the feature interactions through a wide set of cross product feature transformation effectively. But for generalization it would require a lot of feature engineering efforts. In case of deep neural networks the generalization can be done better to unseen feature combinations through low-dimensional dense embedding learned for the sparse features with less feature engineering efforts. The drawback of deep neural networks is that it can over generalize and recommend less relevant items when the user-item interactions are sparse and high-rank. In case of wide and deep model, both the wide model and the deep model are jointly trained in order to combine the benefits of memorization and the generalization. The brief structure of wide and deep model can be seen in Fig. 3.4. The aim is to achieve both the memorization and generalization in one single model by jointly training the linear model component and neural network component.

In this study, we passed all the raw categorical features (these were the virus types, HPV, TCT e.t.c) to the wide part and also made two artificial features as required by the model by crossing some the same categorical attributes to introduce some form of non-linearity. The deep part feed forward neural network with three layers that takes as input embedding vectors of the

virus types and also taking the viral load and all continuous features in general as raw inputs. Though this model was implemented for recommendation problem by the team at Google, we considered it may also do well for this research mainly because of its ability to generate good results on problems of sparse matrices. It has also been used in the medical field for predicting the onset of type 2 diabetes electronic health records [32] and classification of chronic kidney disease [33]. In the wide set the feature set includes the raw input features and transformed features. They use cross product transformation for the transformed features. This captures the interactions between the binary features, and adds non-linearity to the generalized linear model. The deep component is a feed-forward neural network. For categorical features, the original inputs are feature strings. Each of these sparse, high-dimensional categorical features are first converted into a low-dimensional and dense real-valued vector, often referred to as an embedding vector. The embedding vectors are initialized randomly and then the values are trained to minimize the final loss function during model training. These low-dimensional dense embedding vectors are then fed into the hidden layers of a neural network in the forward pass.

Joint training of a Wide and Deep Model is done by back propagating the gradients from the output to both the wide and deep part of the model simultaneously using mini-batch stochastic optimization. Joint training optimizes all parameters simultaneously by taking both the wide and deep part as well as the weights of their sum into account at training time. For joint training the wide part only needs to complement the weaknesses of the deep part with a small number of cross-product feature transformations, rather than a full-size wide model.

This model is open sourced in Tensorflow by google which we have used in our project for predicting the results.

The model architecture for this task that produced the best metrics after several hyper-parameter tuning is given below (this architecture represents just the hidden layer of the deep part, the inputs and the single neuron output layer has been excluded):

The wide part is just a simple linear model that approximates the function

$$y = \beta + \sum_{i=1} w_i * x_i$$

Table 3.3: Model Architecture for the Deep part

Layer (type)	Output Shape
Dense	(None, 16)
Dense	(None, 128)
Dense	(None, 32)

3.4.2 MLP

A Multi-layered Perceptron(MLP) is a type of neural network that can have one or more layers, these models can be broadly categorised as a deep model. MLP's works well for datasets with large number of training examples with dense matrices, they can over-generalise for sparse and high rank matrix leading to inaccurate predictions and this is why the wide and deep model tries to check for this flaw by incorporating a linear model as earlier stated in the previous section. For this project we used a 3-layer(12,8,1) MLP where the all layers used the "Relu" activation function except the output layer which used "Sigmoid function" because we wanted a binary output. The loss function for this network was "Binary Cross Entropy" while the optimizer used was the "Adam" optimizer.

It is important to note that we tried various configurations for the MLP, changing the number of layers and their number of neurons, activation functions, optimizers, and loss functions, but this framework produced better results amongst the configurations we tried.

3.4.3 Random Forest

Random Forest has been used extensively for cervical cancer detection post-ing good results in comparison to other machine learning models. they operate by creating several decision trees and outputting the mean of the predictions of all trees as its final prediction. The model was trained using SciKit learn Random forest library with an added step of performing grid search CV to determine the number of trees and the max depth to select a framework that gives the best recall.

3.4.4 SVM

Since the Wide and Deep Model is a combination of both a linear model and a deep model, we decided to also compare its performance with that of a linear model, this and the fact that Support Vector Machines(SVM) has been used in a significant amount of existing works on cervical cancer detection guided our choice to also see how Support Vector Machines performs on the dataset. The Scikit learn library was used for training the SVM with a linear kernel.

3.5 Implementation

The default parameters of the wide&deep model were kept same as in [28]. We implemented the MLP using Keras while SVM and Random Forest using Scikit learn libraries. We applied ten fold Cross-Validation during the training. The selected models were evaluated on five performance metrics : Recall(REC), Specificity(SPE), Precision(PRE), F1-score(F1), and Balanced Accuracy(BER).

3.6 Class Imbalance

For the imbalanced-class issue, the dataset was augmented by oversampling the minority class using the Synthetic Minority Oversampling Technique(SMOTE) [34]. This method has been proven to provide better results and is less susceptible to over-fitting compared to the fundamental method that duplicates data points of the minority class repeatedly.

The authors steered clear of the under-sampling approach because of how delicate the dataset is. It was essential and counter-intuitive not to lose information from any patient as under-sampling would shrink the dataset from 5973 to 1664.

When up-sampling the data and cross-validating, we were cautious with this process not to expose the model to examples of the test set. We made sure of this by always taking out the validation set before up-sampling the remaining nine partitions for every fold of the cross-validation process. This way, we ensure that the model does not know the test data. We also do not test the model with synthetic data points.



Figure 3.5: Over-Sampling and Cross-Validation strategy [35]

Chapter 4

Deployment

In order to provide end user support a web application has been designed where a user can enter the test details and get the predicted value for Pathology whether the patient has cancer or not with the confidence value.

The web application is has a Python Flask framework. As the link is accessed a login page will be presented, after entering the login credentials, it will present with the homepage. One can check the details of the page and after clicking the test page it will present with the details form designed using HTML and after entering the details and clicking the Predict button, the prediction results will be given in the results page which gives the Pathology value and the Confidence of the prediction.

The model has been deployed in the Google Cloud Platform(GCP). GCP provides with a tool called App engine which is a Platform as a Service(PaaS) where the application code and data needs to be uploaded and the rest of the resources will be allocated by GCP. The scaling will also be maintained by GCP as the input requests increase.

Chapter 5

Results

As mentioned above, we evaluated two kinds of feature combinations: D_1): Raw dataset after preprocessing (39 features); D_2): Hand-picked dataset (19 features); D_3): Up-sampled dataset of D_1 (39 features); D_4): Up-sampled dataset of D_2 (19 features).

As can be seen from Table.5, on the original dataset(D_1), the Wide&Deep model performed better than every other model on REC(65%) and SPEC(96%), which is vital for the domain of cervical cancer prediction. At the same time, RF shows a lower REC and SPE compared to the Wide&Deep model. The results for MLP and SVM were the least scores. These results may demonstrate that the Wide&Deep model has learned more patterns than other models with full features.

For the hand-picked dataset (D_2), which was informed majorly by medical information of the chief contributor viruses to the presence of CIN, the MLP model performed significantly better than the others with REC(60%) and SPE(97%), respectively. We could assume that a smaller set of feature set has adverse effects on the Wide and Deep model, while it favors the Random Forest model.

For the up-sampled dataset(D_3) which was originated from dataset(D_1), the Random-Forest and Wide and Deep are at par given balanced accuracy, although the former performed better on the positive class(minority class). Simultaneously, the latter performs better on the negative class and more precise with its positive predictions. The SMOTE method enhanced the overall performance for each model. Compared to D_1 , the BER was increased from 81% to 89% in RF, the value for REC has been increased from D1 best model 65% to D_3 best model 90%. This result shows that the upsampling of

imbalanced data could help us gain a higher performance overall models.

Lastly, the up-sampled dataset(D_4) was derived from hand-picked dataset(D_2) with SMOTE. From tabel.5, we could see the Deep&Wide model achieved the best overall-balanced performance with REC 92%, SPC 88%, and BER 90%. While the RF demonstrated an excellent and stable performance compared with other two models with REC 95%, SPC 82%, and BER 89%. We also observed that the REC in D_4 was averagely higher than D_3 while the SPE was lower.

Table 5.1: Final performance of models

Model	D_1 : Raw data				
	REC	SPE	PRE	$F1$	BER
Wide and Deep	65	96	68	67	81
RF	62	97	71	66	79
MLP	56	97	65	60	77
SVM	49	97	72	58	73
Model	D_2 : Hand-picked/Chi2 data				
	REC	SPE	PRE	$F1$	BER
Wide and Deep	57	97	71	63	77
RF	57	97	74	64	77
MLP	60	97	69	64	79
SVM	53	97	66	59	75
Model	D_3 :Raw data with SMOTE				
	REC	SPE	PRE	$F1$	BER
Wide and Deep	87	90	53	66	89
RF	90	88	49	63	89
MLP	81	92	55	65	87
SVM	74	94	76	75	84
Model	D_4 : Hand-picked/chi data with SMOTE				
	REC	SPE	PRE	$F1$	BER
Wide and Deep	92	88	47	62	90
RF	95	82	40	56	89
MLP	87	89	60	71	88
SVM	93	81	37	53	87

Chapter 6

Discussion

Early diagnosis of cervical cancer is very important clinically, but excessive referral and colposcopy test not only wastes medical resources but also easily causes mental stress to patients. The main purpose of this research is to build a clinical prediction system to assist doctors in diagnosis with the help of machine learning, so as to reduce the excessive use of colposcopy as much as possible. Based on the records of 5973 outpatients, we trained a decision-making assistance system using Wide&Deep learning with a 92% sensitivity, and 88% specificity.

The final performance for REC and SPE are quite essential for clinical decision support system. Given the results in this paper, we argued that getting more instances especially of the positive class(CIN2+) will be helpful for the model as we will eliminate the need to up-sample the data, but this is usually not an easy task to accomplish considering the plethora of issues involved with gathering medical data. The Wide&Deep model and Random Forest model for the up-sampled dataset can be deployed and further improved by online learning on every new instances evaluated by the model.

The Wide&Deep model reacts negatively from the reduction of the feature set, as can be seen from the drastic reduction performance for the hand-picked dataset(D_2) and also the chi-square selected top 15 features. However, the performance of wide&deep mode gained the best performance on the up-sampling hand-picked Dataset(D_4). The reduction in the number of features will locally affect the performance of the wide&deep, but is not the most fatal. The size of data is a crucial factor that affects the model performance, especially the number of occurrences of CIN2+ as also described above.

The final result(D_4) shows that feature selection can promote the im-

provement of model sensitivity. However, considering the average specificity, we find that the increase in sensitivity comes at the cost of reducing the specificity, which means that there are more of non-cancer patients are predicted to be cancer patients. This trade-off is also in a clinical dilemma. We believe that more positive samples can improve the current situation.

Chapter 7

Conclusion

This paper evaluated the Wide&Deep learning model with SMOTE to train a classifier that supports an accurate triage during the cervical cancer screening process, given a total of 5973 consecutive outpatients undergoing cervical cancer screening followed by colposcopy testing. Feature selection has been made based on the domain knowledge, Visualization, and Chi-square test. The final result shows that the Wide&Deep model has shown the best performance on up-sampled data. The performance may benefit from the increased size of positive cases. The current model could be deployed with a web page portal in a user-friendly interface for clinical usage in the future

Acknowledgement

Deeksha Ramakrishna, Stanley Umeh contribute this paper equally.

Bibliography

- [1] A. Ostör, “Natural history of cervical intraepithelial neoplasia: a critical review.” *International journal of gynecological pathology: official journal of the International Society of Gynecological Pathologists*, vol. 12, no. 2, pp. 186–192, 1993.
- [2] M. F. Mitchell, W. N. Hittelman, W. K. Hong, R. Lotan, and D. Schottenfeld, “The natural history of cervical intraepithelial neoplasia: an argument for intermediate endpoint biomarkers.” *Cancer Epidemiology and Prevention Biomarkers*, vol. 3, no. 7, pp. 619–626, 1994.
- [3] C. Buckley, E. Butler, and H. Fox, “Cervical intraepithelial neoplasia.” *Journal of clinical pathology*, vol. 35, no. 1, pp. 1–13, 1982.
- [4] M. Schiffman, P. E. Castle, J. Jeronimo, A. C. Rodriguez, and S. Wacholder, “Human papillomavirus and cervical cancer,” *The Lancet*, vol. 370, no. 9590, pp. 890–907, 2007.
- [5] A. C. Rodríguez, M. Schiffman, R. Herrero, A. Hildesheim, C. Bratti, M. E. Sherman, D. Solomon, D. Guillén, M. Alfaro, J. Morales *et al.*, “Longitudinal study of human papillomavirus persistence and cervical intraepithelial neoplasia grade 2/3: critical role of duration of infection,” *Journal of the National Cancer Institute*, vol. 102, no. 5, pp. 315–324, 2010.
- [6] Z. Hu and D. Ma, “The precision prevention and therapy of hpv-related cervical cancer: new concepts and clinical implications,” *Cancer medicine*, vol. 7, no. 10, pp. 5217–5236, 2018.
- [7] G. Ronco, S. Franceschi, and N. Segnan, “Hpv16 and hpv18 genotyping in cervical cancer screening.” *The Lancet. Oncology*, vol. 12, no. 9, pp. 831–832, 2011.

- [8] “Human papillomavirus (hpv) and cervical cancer,” [https://www.who.int/news-room/fact-sheets/detail/human-papillomavirus-\(hpv\)-and-cervical-cancer](https://www.who.int/news-room/fact-sheets/detail/human-papillomavirus-(hpv)-and-cervical-cancer).
- [9] I. Kalliala, A. Athanasiou, A. Veroniki, G. Salanti, O. Efthimiou, N. Raftis, S. Bowden, M. Paraskevaïdi, K. Aro, M. Arbyn *et al.*, “Incidence and mortality from cervical cancer and other malignancies after treatment of cervical intraepithelial neoplasia: a systematic review and meta-analysis of the literature,” *Annals of oncology*, vol. 31, no. 2, pp. 213–227, 2020.
- [10] F. Asadi, C. Salehnasab, and L. Ajori, “Supervised algorithms of machine learning for the prediction of cervical cancer,” *J. Biomed. Phys. Eng.*, vol. 10, no. 4, pp. 513–522, 2020.
- [11] A. A. Abdullah, N. K. Abu Sabri, W. Khairunizam, I. Zunaidi, Z. M. Razlan, and A. B. Shahrman, “Development of predictive models for cervical cancer based on gene expression profiling data,” *IOP Conf. Ser. Mater. Sci. Eng.*, vol. 557, no. 1, 2019.
- [12] Y. Singh, D. Srivastava, P. S. Chandranand, and S. Singh, “Algorithms for screening of cervical cancer: A chronological review,” *arXiv*, 2018.
- [13] A. Chirkina, M. Medvedeva, and E. Komotskiy, “CIN classification and prediction using machine learning methods,” *AIP Conf. Proc.*, vol. 1836, no. June 2017, 2017.
- [14] S. Gordon, G. Zimmerman, and H. Greenspan, “Image segmentation of uterine cervix images for indexing in pacs,” in *Proceedings. 17th IEEE Symposium on Computer-Based Medical Systems*. IEEE, 2004, p. 298.
- [15] T. Xu, H. Zhang, X. Huang, S. Zhang, and D. N. Metaxas, “Multimodal deep learning for cervical dysplasia diagnosis,” in *International conference on medical image computing and computer-assisted intervention*. Springer, 2016, pp. 115–123.
- [16] R. Zhang, Y. Zheng, T. W. C. Mak, R. Yu, S. H. Wong, J. Y. Lau, and C. C. Poon, “Automatic detection and classification of colorectal polyps by transferring low-level cnn features from nonmedical domain,” *IEEE journal of biomedical and health informatics*, vol. 21, no. 1, pp. 41–47, 2016.

- [17] J. Yessi, S. Ng, and N. Osman, “Intelligent screening systems for cervical cancer,” *The Scientific World Journal* 2014, p. 810368, 2018.
- [18] K. Ravikumar, K. L. MacLaughlin, M. R. Scheitel, M. Kessler, K. B. Waghlikar, H. Liu, and R. Chaudhry, “Improving the accuracy of a clinical decision support system for cervical cancer screening and surveillance,” *Applied clinical informatics*, vol. 9, no. 1, p. 62, 2018.
- [19] S. Y. M. Ahmed, S. M. Freire, T. M. P. Feitosa, L. M. G. Zardo, and R. T. d. Almeida, “Ad-siscolo: a decision-support tool to aid the management of a cervical cancer screening program,” *Research on Biomedical Engineering*, no. AHEAD, pp. 0–0, 2018.
- [20] P. Bountris, M. Haritou, A. Pouliakis, N. Margari, M. Kyrgiou, A. Spathis, A. Pappas, I. Panayiotides, E. A. Paraskevaidis, P. Karakitsos *et al.*, “An intelligent clinical decision support system for patient-specific predictions to improve cervical intraepithelial neoplasia detection,” *BioMed research international*, vol. 2014, 2014.
- [21] P. Bountris, M. Haritou, A. Pouliakis, P. Karakitsos, and D. Koutsouris, “A decision support system based on an ensemble of random forests for improving the management of women with abnormal findings at cervical cancer screening,” in *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2015, pp. 8151–8156.
- [22] M. Kyrgiou, A. Pouliakis, J. G. Panayiotides, N. Margari, P. Bountris, G. Valasoulis, M. Paraskevaidi, E. Bilirakis, M. Nasioutziki, A. Loufopoulos *et al.*, “Personalised management of women with cervical abnormalities using a clinical decision support scoring system,” *Gynecologic Oncology*, vol. 141, no. 1, pp. 29–35, 2016.
- [23] M. Berggrund, I. Gustavsson, R. Aarnio, J. Hedlund-Lindberg, K. Sanner, I. Wikström, S. Enroth, M. Olovsson, and U. Gyllensten, “Hpv viral load in self-collected vaginal fluid samples as predictor for presence of cervical intraepithelial neoplasia,” *Virology journal*, vol. 16, no. 1, pp. 1–10, 2019.
- [24] C.-A. Sun, H.-C. Lai, C.-C. Chang, S. Neih, C.-P. Yu, and T.-Y. Chu, “The significance of human papillomavirus viral load in prediction of

histologic severity and size of squamous intraepithelial lesions of uterine cervix,” *Gynecologic oncology*, vol. 83, no. 1, pp. 95–99, 2001.

- [25] L. Del Río-Ospina, S. C. Soto-De León, M. Camargo, D. A. Moreno-Pérez, R. Sánchez, A. Pérez-Prados, M. E. Patarroyo, and M. A. Patarroyo, “The dna load of six high-risk human papillomavirus types and its association with cervical lesions,” *BMC cancer*, vol. 15, no. 1, pp. 1–11, 2015.
- [26] J. Kim, B. K. Kim, D. Jeon, C. H. Lee, J.-W. Roh, J.-Y. Kim, and S.-Y. Park, “Type-specific viral load and physical state of hpv type 16, 18, and 58 as diagnostic biomarkers for high-grade squamous intraepithelial lesions or cervical cancer,” *Cancer research and treatment: official journal of Korean Cancer Association*, vol. 52, no. 2, p. 396, 2020.
- [27] C. Constandinou-Williams, S. I. Collins, S. Roberts, L. S. Young, C. B. Woodman, and P. G. Murray, “Is human papillomavirus viral load a clinically useful predictive marker? a longitudinal study,” *Cancer Epidemiology and Prevention Biomarkers*, vol. 19, no. 3, pp. 832–837, 2010.
- [28] H. T. Cheng, L. Koc, J. Harmsen, T. Shaked, T. Chandra, H. Aradhye, G. Anderson, G. Corrado, W. Chai, M. Ispir, R. Anil, Z. Haque, L. Hong, V. Jain, X. Liu, and H. Shah, “Wide & deep learning for recommender systems,” *ACM Int. Conf. Proceeding Ser.*, vol. 15-September-2016, pp. 7–10, 2016.
- [29] N. Chawla, K. Bowyer, L. Hall, and W. Kegelmeyer, “Smote: Synthetic minority over-sampling technique. jair 16,” *Link: <https://bit.ly/3au15Za>*, 2002.
- [30] A. Marley. (2018) TeachMeObGyn. [Online]. Available: <https://teachmeobgyn.com/gynaecology/cervix/cervical-cancer/>
- [31] V. Mello and R. K. Sundstrom, “Cervical intraepithelial neoplasia (cin),” *StatPearls [Internet]*, 2020.
- [32] B. P. Nguyen, H. N. Pham, H. Tran, N. Nghiem, Q. H. Nguyen, T. T. Do, C. T. Tran, and C. R. Simpson, “Predicting the onset of type 2 diabetes using wide and deep learning with electronic health records,” *Computer methods and programs in biomedicine*, vol. 182, p. 105055, 2019.

- [33] A. Al Imran, M. N. Amin, and F. T. Johora, “Classification of chronic kidney disease using logistic regression, feedforward neural network and wide & deep learning,” in *2018 International Conference on Innovation in Engineering and Technology (ICIET)*. IEEE, 2018, pp. 1–6.
- [34] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *J. Artif. Intell. Res.*, vol. 16, no. February 2017, pp. 321–357, 2002.
- [35] L. Maddalena, I. Granata, I. Manipur, M. Manzo, and M. R. Guarracino, “Glioma grade classification via omics imaging,” *BIOIMAGING 2020 - 7th Int. Conf. Bioimaging, Proceedings; Part 13th Int. Jt. Conf. Biomed. Eng. Syst. Technol. BIOSTEC 2020*, no. January, pp. 82–92, 2020.

List of Figures

1.1	Types of CIN [30]	3
3.1	Correlation Matrix	6
3.2	Class Distribution	7
3.3	Positive risk with virus type distribution	8
3.4	The spectrum of wide and deep model [28]	10
3.5	Over-Sampling and Cross-Validation strategy [35]	14

List of Tables

3.1	Feature table	8
3.2	Hand-picked and Chi-square feature table	9
3.3	Model Architecture for the Deep part	12
5.1	Final performance of models	18